

AER1503 Project Progress Report

Optimizing transfer trajectories for Earth-Mars interplanetary orbital maneuvers

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I. PROJECT OBJECTIVE

The primary goal of this project is to develop a comprehensive Python-based optimization framework that identifies and optimizes interplanetary transfer trajectories between Earth and Mars. Using the OpenMDAO Python framework, the software will consider several design variables in order to describe an optimal trajectory to Mars. Since OpenMDAO is a multidisciplinary optimization (MDO) framework, the project will take advantage of its capabilities by optimizing several subsystems (i.e orbital mechanics, propulsion, thermal, and power) separately, which will then be combined into a larger global environment.

In the project's current phase, we have formed a case study optimization problem that seeks to minimize the delta-V used in a one way Earth to Mars mission, using the method of patched conics. Formally, we solve

$$\text{minimize } \Delta V \quad (1)$$

$$\text{subject to } g_i(x), \quad i = 1, 2, \dots, n \quad (2)$$

where $g_i(x)$ are constraints imposed on the system [1].

For the interim, we have not constrained the model, as the focus was to simply obtain a working and somewhat realistic model of an interplanetary mission. In the final project, we hope to impose constraints on the time of flight and maximum allowable delta-V, in order to obtain a more feasible solution.

We also only considered the radii of Low Earth Orbit (r_{LEO}) and Low Mars Orbit (r_{LMO}) as design variables for the model, with restrictions being [2]

$$6538.0 \text{ km} \leq r_{LEO} \leq 8378.0 \text{ km} \quad (3)$$

$$3490.0 \text{ km} \leq r_{LMO} \leq 3590.0 \text{ km} \quad (4)$$

We anticipate considering more design variables in the project's final iteration, particularly the mission's time of flight and mass of the propulsion used, among others.

II. CASE STUDY: ONE WAY EARTH-MARS HOHMANN TRANSFER

For the project's preliminary phase, we consider a simple, coplanar, Hohmann transfer between circular Low Earth Orbit (LEO) to circular Low Mars Orbit (LMO). Despite the model's relative simplicity, the assumption of a coplanar Hohmann transfer is justified, as the difference in inclination (relative to the ecliptic plane) between the orbits of Earth and Mars is

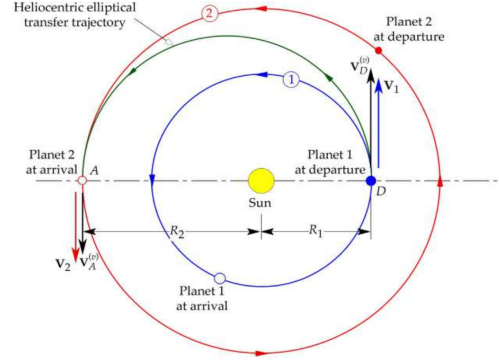


Fig. 1. A Hohmann transfer from inner planet 1 (Earth) to outer planet 2 (Mars). This image is taken from Chapter 8 of Howard Curtis' *Orbital Mechanics for Engineering Students* [3].

about 1.85° [3]. To model both an orbit change and rendezvous with Mars upon arrival, we use the method of patched conics and split the trajectory into three distinct parts (and subsequent frames of reference).

First, we assume that a spacecraft starts out in LEO ($r_{LEO} = 6778 \text{ km}$), and we create an OpenMDAO subsystem (EarthOrbitComp) that computes its circular orbital velocity v_{LEO} . Another subsystem (EarthEscapeDVComp) then computes the delta-V ΔV_E required for the spacecraft to depart LEO and enter a heliocentric transfer orbit (along with the hyperbolic excess speed $v_{\infty,E}$).

Next, the spacecraft enters a heliocentric transfer orbit, where it coasts until it is "captured" by Mars' sphere of influence. OpenMDAO subsystems for this phase compute the speed when the spacecraft enters the heliocentric orbit and the speed when it arrives at Mars' orbit (HelioTransferOrbitComp). A subsystem HelioInjectionMatchComp compares the hyperbolic excess speed $v_{\infty,E}$ calculated by EarthEscapeDVComp with that calculated by HelioInjectionMatchComp to determine which is more realistic.

Finally, the MarsOrbitComp subsystem calculates the circular orbital velocity needed by the spacecraft to stay in LMO, while the MarsInjectionMatchComp subsystem computes the hyperbolic excess speed $v_{\infty,M}$ using the heliocentric arrival speed. Then, the MarsCaptureDVComp subsystem computes the delta-V ΔV_M needed to capture the

spacecraft into Mars orbit.

The total delta-V ΔV needed for this mission is then described by

$$\Delta V = \Delta V_E + \Delta V_M \quad (5)$$

which is what we seek to minimize.

A more detailed outline of the OpenMDAO framework, along with a rundown of the various subsystems used in the project model, will be provided in the final project report.

III. CASE STUDY RESULTS

After applying the patched-conic model to the Hohmann transfer scenario described by Figure 1 using the OpenMDAO framework, we obtained results that seem feasible.

Prior to the optimization, we first ran the model to calculate a non-optimal delta-V, using equations of orbital mechanics described in Chapter 8 of the course's main textbook. In doing so, we obtained a delta-V of about 2.254 km/s, with a flight time of about 259 days. The model's time of flight agrees with the results provided for the transfer time t_{12} from Slide 7, Chapter 10 of the Course Notes, suggesting that the model has been implemented correctly.

Using the `run_driver` method of the OpenMDAO `Problem` class, we then proceeded to optimize the results of the previously run model. Now, given that the Hohmann transfer is commonly regarded as "the most energy-efficient approach for a spacecraft to transfer from one planet's orbit to another", we would expect the optimal delta-V to be close to that of the non-optimal delta-V (2.254 km/s). Upon running the OpenMDAO optimizer (which is gradient descent based), we obtain an optimal delta-V of about 2.241 km/s, which is indeed close to the non-optimal delta-V previously obtained.

This preliminary test confirms our correct implementation of the problem defined in Sections I and II, and sets the stage for us to consider more complicated trajectories or mission parameters in the final version.

IV. DEVIATIONS FROM PROJECT DEFINITION

In our initial Project Definition, the goal was to develop "a comprehensive Python-based software that identifies and optimizes interplanetary transfer trajectories between Earth and Mars". At this interim stage of the project, we can safely state that this goal has been largely followed, with the exception of a few deviations.

Notably, our current version of the project considers only a coplanar Hohmann transfer for interplanetary missions, instead of the low-thrust and gravity-assisted maneuvers mentioned in the Project Definition. However, as our overall goal is to minimize the delta-V used in these kinds of missions, the current implementation might be satisfactory.

In addition, we have not implemented subsystems for thermal and propulsion to date. Given OpenMDAO's limitation in only being able to optimize one objective function at a time, our focus remains to optimize the delta-V required in these missions, as that is often the most important factor in trajectory-related decisions.

V. NEXT STEPS

Following the successful initial implementation of our rudimentary model of a one-way Earth-Mars mission, the obvious next step is to implement a return trip from Mars to Earth. Additional OpenMDAO subsystems will need to be developed for this, but will mostly follow the style of the existing code.

We would also like to implement perturbations that the spacecraft might experience during flight, as learned in class. These include perturbative effects such as gravitational perturbations and solar radiation pressure.

Time permitting, other steps will be explored for implementation as well, in consultation with the course instructor.

REFERENCES

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- [3] H. D. Curtis, *Orbital Mechanics for Engineering Students*, 4th ed., ser. Elsevier aerospace engineering series. San Diego: Elsevier Science & Technology, 2019.